Linear independence properties of the signature components of time-augmented stochastic processes.

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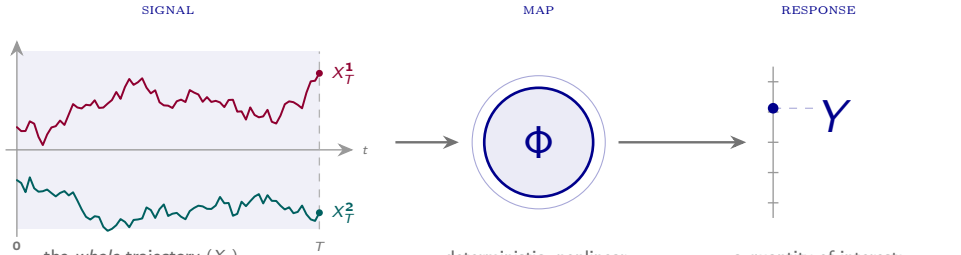




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Motivation



$$Y \;=\; \Phi\big((X_t)_{t\in[0,T]}\big)$$

Goal. Build an approximation $\tilde{\Phi}$ of Φ that is cheap to evaluate and tractable to fit.

The signature of a path

For a path of bounded variation $X := (X_t^1, \cdots, X_t^d)_{t\in[0,T]}$, and for each word $i_1 \cdots i_n$ with $i_k \in \{0, \cdots, d\}$ and $n \in \mathbb{N}^*$ a positive integer, we define:

$$\forall t \in [0,T], \quad \widehat{\mathbb{X}}_t^{i_1\cdots i_n} = \int_0^t \int_0^{t_1} \cdots \int_0^{t_{n-1}} \mathrm{d}\widehat{X}_{t_n}^{i_1} \cdots \mathrm{d}\widehat{X}_{t_1}^{i_n},$$

and

$$\forall t \in [0,T], \quad \widehat{\mathbb{X}}_t^{\emptyset} = 1,$$

where $\emptyset$ denotes the empty word, $\widehat{X}_t^0 = t$ and $\widehat{X}_t^i = X_t^i$ for $i \in \{1, \cdots, d\}$.

- The time-augmented signature of X is $\widehat{\mathbb{X}} := (\widehat{\mathbb{X}}^{i_1\cdots i_n})_{(i_1,\cdots,i_n)\in\{0,\cdots,d\}^n, n\in\mathbb{N}}$.
- For $N \in \mathbb{N}$, the N-step truncated time-augmented signature of X is $\widehat{\mathbb{X}}^{\leq N} := (\widehat{\mathbb{X}}^{i_1\cdots i_n})_{(i_1,\cdots,i_n)\in\mathcal{A}^n, n\leq N}$
  $\rightsquigarrow \forall t \in [0,T], \quad \widehat{\mathbb{X}}_t^{\leq N} \in \mathbb{R}^{\frac{(d+1)^{N+1}-1}{d}} \rightsquigarrow$ finite family of features.
- Libraries for efficiently computing signatures include: *iisignature* [Reizenstein and Graham, 2020], *signatory*, *esig*.

When X is a continuous semimartingale: we define its time-augmented signature using Stratonovich integration theory.

Linear functionals on the time-augmented signature terminal value are universal approximators. [Cuchiero et al., 2026, Bayer et al., 2025a, Bayer et al., 2025b].

$$\Phi(X) \simeq \underbrace{\langle \beta, \widehat{\mathbb{X}}_T^{\leq N} \rangle}_{\tilde{\Phi}(X)}, \quad \text{for large } N \text{ and some } \beta \in \mathbb{R}^{\frac{(d+1)^{N+1}-1}{d}}$$

nonlinear path-dependent functional $\rightsquigarrow$ linear model in nonlinear path features

$$\beta_N^* := \arg\min_{\beta} \mathbb{E}[|\Phi(X) - \langle \beta, \widehat{\mathbb{X}}_T^{\leq N} \rangle|^2]$$

Finding a basis of the linear span of the components of $\widehat{\mathbb{X}}_T^{\leq N}$ is interesting for several reasons :

- β_N^* is unique $\iff$ components of $\widehat{\mathbb{X}}_T^{\leq N}$ are linearly independent $\rightsquigarrow$ **improves interpretability**
- reduction of **computational and storage cost of features are reduced**
- the **prediction performance is preserved**, while empirical improvements may arise in practice due to improved conditioning, reduced variance, or more stable cross-validation.

Notations.

- $\mathcal{W}_0 = \{\emptyset\}$,
- $\mathcal{W}_N = \{i_1 i_2 \cdots i_N \mid (i_1, \cdots, i_N) \in \mathcal{A}^N\}$ for $N \in \mathbb{N}^* = \mathbb{N} \setminus \{0\}$, $\rightsquigarrow$ set of words of length N
- $\mathcal{W}_{\leq N} = \bigsqcup_{k=0}^{N} \mathcal{W}_k$ for $N \in \mathbb{N}$, $\rightsquigarrow$ set of words of length not greater than N,
- $\mathcal{W}_{<\infty} = \bigsqcup_{k=0}^{\infty} \mathcal{W}_k$, $\rightsquigarrow$ set of words of finite length

We denote the length of a word $w \in \mathcal{W}_{<\infty}$ by $\|w\|$.

Definition (Shuffle product)

The shuffle product between two words $\sqcup\!\sqcup : \mathcal{W}_{<\infty} \times \mathcal{W}_{<\infty} \longrightarrow \mathrm{Span}(\mathcal{W}_{<\infty})$ is defined inductively for all $w, v \in \mathcal{W}_{<\infty}$ and $i, j \in \mathcal{A}$ as follows :

$$\begin{cases} w \sqcup\!\sqcup \emptyset = \emptyset \sqcup\!\sqcup w = w \\ (wi) \sqcup\!\sqcup (vj) = (w \sqcup\!\sqcup (vj))i + ((wi) \sqcup\!\sqcup v)j. \end{cases}$$

The shuffle product can be extended to $\mathrm{Span}(\mathcal{W}_{<\infty}) \times \mathrm{Span}(\mathcal{W}_{<\infty})$ by bilinearity.

Remark: We can easily observe that the words which appear in the shuffle of w and v all have length $\|w\| + \|v\|$.

Example: $12 \sqcup\!\sqcup 01 = 1201 + 1021 + 0121 + 1012 + 0112 + 0112$.

Proposition (Shuffle product property)

Let $X : [0, T] \to \mathbb{R}^d$ be a path such that its time-augmented signature $\widehat{\mathbb{X}}$ is well defined. Then

$$\forall t \in [0, T], \forall \boldsymbol{w}, \boldsymbol{v} \in \mathcal{W}_{<\infty}, \quad \langle \boldsymbol{w}, \widehat{\mathbb{X}}_t \rangle \langle \boldsymbol{v}, \widehat{\mathbb{X}}_t \rangle = \langle \boldsymbol{w} \sqcup\!\sqcup \boldsymbol{v}, \widehat{\mathbb{X}}_t \rangle.$$

Example: Let us consider a path $X := (X^1, X^2) : [0, T] \to \mathbb{R}^2$. Hence we have
$\int_0^T dX_t^1 \int_0^T dX_t^2 = \widehat{\mathbb{X}}_T^1 \widehat{\mathbb{X}}_T^2 = \langle 1, \widehat{\mathbb{X}}_T \rangle \langle 2, \widehat{\mathbb{X}}_T \rangle = \langle 1 \sqcup\!\sqcup 2, \widehat{\mathbb{X}}_T \rangle = \langle 12 + 21, \widehat{\mathbb{X}}_T \rangle = \int_0^T \int_0^t dX_s^1 dX_t^2 + \int_0^T \int_0^t dX_s^2 dX_t^1$.

Generalization of the integration by parts formula.

$(\Omega, \mathcal{F}, \mathbb{P})$ probability space and $X = (X_t)_{t\in[0,T]}$ be a continuous stochastic process of law $\mathbb{P}_X$ such that its time-augmented signature is well defined.

We denote the word $\underbrace{00\cdots0}_{k}$ by 0_k for $k \in \mathbb{N}$. Since $\langle 0_k, \widehat{\mathbb{X}}_T\rangle = \frac{T^k}{k!} > 0$, the shuffle product property yields

$$\forall k \in \mathbb{N}, \quad \langle \boldsymbol{w}, \widehat{\mathbb{X}}_T\rangle = \langle \frac{k!}{T^k}(\boldsymbol{w} \sqcup\!\sqcup 0_k), \widehat{\mathbb{X}}_T\rangle,$$

$$\frac{k!}{T^k}(\boldsymbol{w} \sqcup\!\sqcup 0_k) \rightsquigarrow \text{linear combination of words of length } |||w||| + k.$$

Consequence: For $N \in \mathbb{N}$ and $\boldsymbol{w} \in \mathcal{W}_{\leq N} \setminus \mathcal{W}_N$. Taking $k = N - |||\boldsymbol{w}||| > 0$, we deduce

$$\widehat{\mathbb{X}}_T^{\boldsymbol{w}} = \text{linear combination of components associated with words of length } N.$$

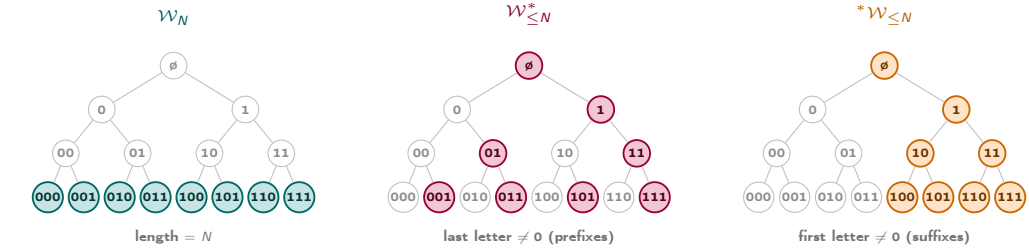
The signature of a path : spanning results

Theorem

$Span(\widehat{\mathbb{X}}_T^{\boldsymbol{w}} \mid \boldsymbol{w} \in \mathcal{W}_N) = Span(\widehat{\mathbb{X}}_T^{\boldsymbol{w}} \mid \boldsymbol{w} \in \mathcal{W}_{\leq N})$, whatever the law $\mathbb{P}_X$.

Other subsets of $\mathcal{W}_{\leq N}$ than $\mathcal{W}_N$ share the same spanning property, such as

- $\mathcal{W}^*_{\leq N} := \{\emptyset\} \cup \{i_1 \cdots i_n \in \mathcal{W}_{\leq N} \mid i_n \neq 0\}$ (prefixes)
- ${}^*\mathcal{W}_{\leq N} := \{\emptyset\} \cup \{i_1 \cdots i_n \in \mathcal{W}_{\leq N} \mid i_1 \neq 0\}$ (suffixes)



All these sets have the same cardinality $(d+1)^N$ (above: $d = 1$, $N = 3 \rightsquigarrow 8$ words out of 15).

Theorem

Let $m, d \in \mathbb{N}^*$, $(B_t)_{t\in[0,T]}$ be a m-dimensional Brownian motion, and Y be a random variable independent of $(B_t)_{t\in[0,T]}$. Let $(X_t)_{t\in[0,T]} := (X_t^1, \cdots, X_t^d)_{t\in[0,T]}$ be the solution to:

$$dX_t = \sigma(t, X_t)dB_t + b(t, X_t)dt, \quad X_0 = Y, \tag{1}$$

where

- $b : [0, T] \times \mathbb{R}^d \to \mathbb{R}^d$ *and* $\sigma : [0, T] \times \mathbb{R}^d \to \mathbb{R}^{d\times m}$ *are* $C^{1,3}$ *with linear growth in space uniformly in time,*
- $\exists \lambda > 0, \forall t \in [0, T], \forall x \in \mathbb{R}^d, \forall \xi \in \mathbb{R}^d, \quad \xi\sigma(t, x)\sigma(t, x)^\top \xi^\top \geq \lambda\|\xi\|^2$.

Then for all $N \in \mathbb{N}$, the family of random variables $(\widehat{\mathbb{X}}_T^{\boldsymbol{w}})_{\boldsymbol{w}\in\mathcal{W}^*_{\leq N}}$ is linearly independent.

The result still holds when considering affine interpolation of the solution on a time grid.

First step:
For $(W_t)_{t\in[0,T]}$ a d-dimensional Brownian motion, $(\widehat{\mathbb{W}}_T^{\boldsymbol{w}})_{\boldsymbol{w}\in\mathcal{W}^*_{\leq N}}$ is linearly independent.

1. The family of stochastic processes $(\widehat{\mathbb{W}}_t^{\leq N})_{t\in[0,T]}$ is linearly independent for the $dt\otimes d\mathbb{P}$-a.e. equality.
2. $\forall \boldsymbol{w}\in\mathcal{W}^*_{\leq N}$, $\mathbb{E}\left[\widehat{\mathbb{W}}_T^{\boldsymbol{w}}|\mathcal{F}_t\right] = \widehat{\mathbb{W}}_t^{\boldsymbol{w}} + \sum_{k=1}^{M^{\boldsymbol{w}}} P_{k,\boldsymbol{w}}(T)\widehat{\mathbb{W}}_t^{\boldsymbol{v}_{k,\boldsymbol{w}}}$, and $\deg(P_{k,\boldsymbol{w}}) + \|\boldsymbol{v}_{k,\boldsymbol{w}}\| \leq \|\boldsymbol{w}\| - 1$.

Second step:
The map $\mathrm{x}\mapsto \sum_{\boldsymbol{w}\in\mathcal{W}^*_{\leq N}} c_{\boldsymbol{w}}\langle \boldsymbol{w}, S_N(\widehat{\mathrm{x}})_T\rangle$ is continuous for the rough-path topology. Hence, for all $\mathrm{x}\in\mathrm{Supp}(\mathrm{X})$, $\sum_{\boldsymbol{w}\in\mathcal{W}^*_{\leq N}} c_{\boldsymbol{w}}\langle \boldsymbol{w}, S_N(\widehat{\mathrm{x}})_T\rangle = 0$. Here $S_N : X\mapsto \widehat{\mathbb{X}}^{\leq N}$.

Third step:
By support theorem, we show that for any piecewise affine path $\gamma:[0,T]\to\mathbb{R}^d$, $S_2(\gamma)\in\mathrm{Supp}(\mathrm{X})$. For W a d-dimensional Brownian motion, we construct $W^{(n)}$ the dyadic piecewise approximation of W. Classical results gives $S_2(W^{(n)})\to \mathrm{W}$ for the rough-path topology.

Efficient computation of signature terms

For each word $\boldsymbol{w} \in \mathcal{W}_{<\infty}$ we denote by $S^{\boldsymbol{w}} : X \mapsto \widehat{\mathbb{X}}_T^{\boldsymbol{w}}$.

Proposition (Time-augmented signature of an affine path)

Let $T > 0$ and $X := (X_t^1, \cdots, X_t^{\boldsymbol{d}})_{t\in[0,T]}$ be an affine path taking values in $\mathbb{R}^d$. Then for all $N \in \mathbb{N}^*$ and for all $\boldsymbol{w} = \boldsymbol{i_1} \cdots \boldsymbol{i_N} \in \mathcal{W}_N$ we have the following equality

$$S^{\boldsymbol{w}}(X) = \frac{1}{N!} \prod_{k=1}^{N} \left(\widehat{X}_T^{\boldsymbol{i_k}} - \widehat{X}_0^{\boldsymbol{i_k}} \right),$$

where $\widehat{X}_t^0 = t$ and $\widehat{X}_t^{\boldsymbol{i}} = X_t^{\boldsymbol{i}}$ for $\boldsymbol{i} \in \{1, \cdots, \boldsymbol{d}\}$.

Definition (Concatenation of paths)

Let $T_1, T_2 > 0$, and $X : [0, T_1] \to \mathbb{R}^d$, $Y : [0, T_2] \to \mathbb{R}^d$ be two paths. We denote by $X * Y : [0, T_1 + T_2] \to \mathbb{R}^d$ the concatenation of X and Y defined as

$$X * Y : t \mapsto \begin{cases} X_t & \text{if } \ t \in [0, T_1] \\ Y_{t-T_1} - Y_0 + X_{T_1} & \text{if } \ t \in [T_1, T_1 + T_2] \end{cases}$$

Proposition (Chen's relation)

Let $T_1, T_2 > 0$, and $X : [0, T_1] \to \mathbb{R}^d$, $Y : [0, T_2] \to \mathbb{R}^d$ be two paths of finite variation. For any word $\boldsymbol{i_1 \cdots i_m}$ we have the following relation

$$S^{\boldsymbol{i_1 \cdots i_m}}(X * Y) = \sum_{k=0}^{m} S^{\boldsymbol{i_1 \cdots i_k}}(X) S^{\boldsymbol{i_{k+1} \cdots i_m}}(Y),$$

with the convention $\boldsymbol{i_1 \cdots i_k} = \boldsymbol{\emptyset}$ if $k = 0$, and $\boldsymbol{i_{k+1} \cdots i_m} = \boldsymbol{\emptyset}$ if $k = m$.

Efficient computation of signature terms

Consider a time horizon $T > 0$, an integer $K \geq 2$, and $X = \Delta X_1 * \cdots * \Delta X_K$ defined as the concatenation of K affine paths $\Delta X_k : [0, \frac{T}{K}] \to \mathbb{R}^d$. We denote by

$$X_{|k} = \Delta X_1 * \cdots * \Delta X_k \text{ : } \textbf{forward concatenation}.$$

For the **forward** approach, we iteratively compute the signature of $X_{|k+1} = X_{|k} * \Delta X_{k+1}$ by applying Chen's relation until we obtain the signature of the full path $X = X_{|K}$. The computation of $S^{i_1 \cdots i_m}(X_{|k+1})$ requires

1. computing $(S^{i_1 \cdots i_j}(X_{|k}))_{j \in \{0, \cdots, m\}}$,
2. computing $(S^{i_{j+1} \cdots i_m}(\Delta X_{k+1}))_{j \in \{0, \cdots, m\}}$,
3. performing $2m$ elementary operations. This corresponds to parsing the $m+1$ elements in the summation, and performing $m-1$ products (since there is no products for $j = 0$ and $j = m$).

Efficient computation of signature terms

We define

- $C(B,k)$ the required number of elementary operations to compute $(S^{\boldsymbol{w}}(X_{|k}))_{\boldsymbol{w}\in B}$,
- $\Phi_{\mathsf{f}}(B) = \{\boldsymbol{w} \in \mathcal{W}_{<\infty} \mid \exists \boldsymbol{v} \in \mathcal{W}_{<\infty}, \boldsymbol{wv} \in B\}$,
- $|||B||| = \sum_{\boldsymbol{w}\in B} \|\boldsymbol{w}\|$.

We obtain the following recursive formula

$$C(B,k) = C(\Phi_{\mathsf{f}}(B), k-1) + 2|||B|||.$$

Since

1. $\Phi_{\mathsf{f}} \circ \Phi_{\mathsf{f}} = \Phi_{\mathsf{f}}$,
2. $C(B,1) = \mathrm{Card}(B)$,

we deduce

$$C(B,k) = \mathrm{Card}(\Phi_{\mathsf{f}}(B)) + 2(k-2)|||\Phi_{\mathsf{f}}(B)||| + 2|||B|||.$$

Total cost formula:

$$C_{\mathsf{f}}(B, K) = \mathrm{Card}(\Phi_{\mathsf{f}}(B)) + 2(K-2)|||\Phi_{\mathsf{f}}(B)||| + 2|||B|||.$$

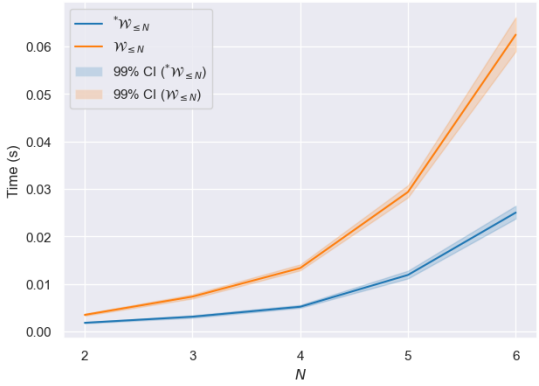
The minimum of $C_{\mathsf{f}}(\cdot, K)$ on the set of subsets of $\mathcal{W}_N$ with cardinality $(d+1)^N$ is attained by ${}^*\mathcal{W}_{\leq N}$. When applied to the set ${}^*\mathcal{W}_{\leq N}$, the gain in elementary operations behaves like

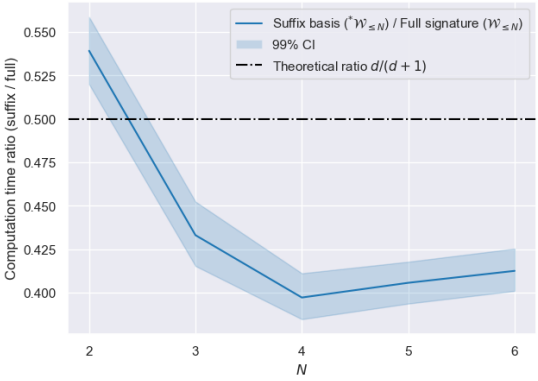
$$\frac{C_{\mathsf{f}}({}^*\mathcal{W}_{\leq N}, K)}{C_{\mathsf{f}}(\mathcal{W}_{\leq N}, K)} \xrightarrow[K\to\infty]{} \frac{d}{d+1}.$$

Numerical results

We used the library `iisignature` and implemented an efficient iterative method to compute suffixes components.

Figure: **Computation time of signature components and comparison with theoretical scaling.** We sample 200 batches of size $n = 500$ of discretized Brownian motion sample paths and compute the empirical mean computation time across batches.





Takeways

- We exhibit sets of words $\mathcal{B} \subset \mathcal{W}_{\leq N}$ s.t.
  1. $\mathrm{Card}(\mathcal{B}) = (d+1)^N$,
  2. $\mathcal{B}$ does not depend on $\mathbb{P}_X$,
  3. $\mathrm{Span}(\widehat{\mathbb{X}}_T^{\boldsymbol{w}} \mid \boldsymbol{w} \in \mathcal{B}) = \mathrm{Span}(\widehat{\mathbb{X}}_T^{\leq N})$.

  $$\Rightarrow \dim\left(\mathrm{Span}\left(\widehat{\mathbb{X}}_T^{\boldsymbol{w}} \mid \boldsymbol{w} \in \mathcal{W}_{\leq N}\right)\right) \leq (d+1)^N, \quad \textbf{whatever } \mathbb{P}_X.$$

- ${}^*\mathcal{W}_{\leq N}$ is such a subset and has the minimal computation cost when using Chen's relation.
- We show that if we consider a process solution to an SDE with non-degenerate diffusion coefficient, then $\dim\left(\mathrm{Span}\left(\widehat{\mathbb{X}}_T^{\boldsymbol{w}} \mid \boldsymbol{w} \in \mathcal{W}_{\leq N}\right)\right) = (d+1)^N$.

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Thanks for your attention!

Questions?